Synopsis Phase 1

Restaurant Food Ordering and Sales System

TEAM MEMBERS:

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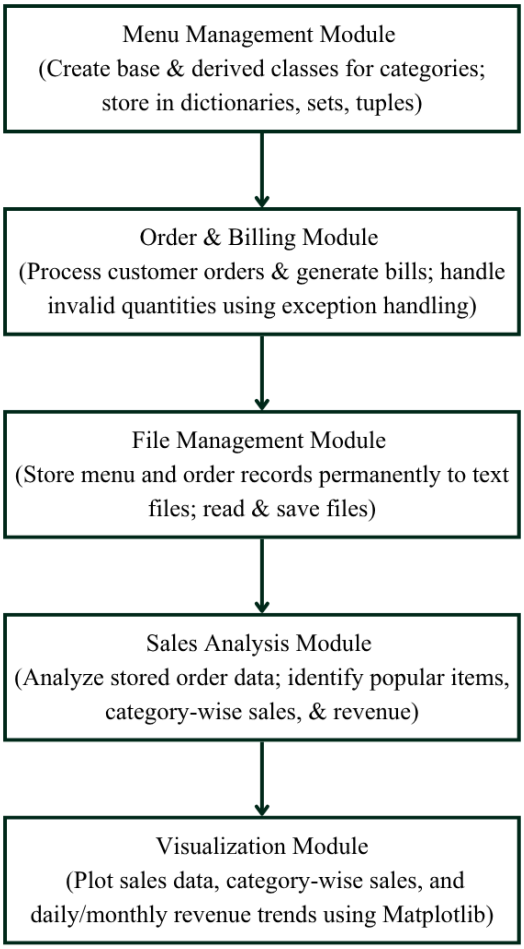
1. **Problem Description:**

The project involves a Python-based restaurant management application for managing food menu items, processing customer orders, and maintaining billing and sales. In reality, the task of manually tracking order details and sales calculations is highly prone to human errors. The application will automate these tasks and ensure the information is permanently saved using files. Additionally, the application will allow categorizing food and managing invalid transactions with exception handling. By analyzing the sales data and plotting the plots using Matplotlib, the manager can effortlessly track the business insights and daily revenue and determine the most popular food items.

2. **Problem Objective**

Design and develop an application to manage food categories, customer orders, and billing information with file handling to store the data permanently and generate visual plots to present the sales reports and insights using Matplotlib.

3. **Steps of Implementation:**



4. **Proposed Modules:**

Module 1 – Menu Management: Create a `FoodItem` base class and derive classes such as `MainCourse`, `Beverage`, and `Dessert`. The classes will implement object-oriented concepts such as inheritance and polymorphism using the `calculate_price()` method.

Module 2 – Order and Billing: Manage the customer orders using lists and dictionaries and generate the bill for the customer orders. The module will also implement exception handling to avoid errors when the customer selects invalid items or quantities.

Module 3 – File Management: Store the menu and order data permanently using text files. Furthermore, the module will read the existing file data in the billing module to ensure it reads the existing menu items and customer orders.

Module 4 – Sales Analysis: Process the permanently stored order data to determine the most popular items and generate revenue in each category and the overall revenue. The module will implement lists, sets, and dictionaries to store and process the data.

Module 5 – Visualization: Plot the most popular food items, sales in each category, and daily or monthly overall revenue using Matplotlib.

5. **All the required topics from the subject:**

Control Structures & Functions: Using the `def` keyword to create functions that carry out individual processes such as the `order_menu`, `read_menu`, and `process_order`. The `match-case` syntax will also be used to implement switch-case statements in selecting the appropriate menu category.

Data Structures: Storing and accessing data using dictionaries for the menu and bills, lists for storing the ordered items, sets to store unique items such as food categories, and tuples for storing fixed information such as food item details.

Exception Handling: Using try-except blocks to catch and handle all errors that may arise when the user provides invalid inputs, such as non-integer values, or when invalid items are selected.

File Handling: Using file-handling functions to read, write, and append new order records to the text files that will permanently store the data.

Object-Oriented Programming (OOP): Creating classes and objects in Python and implementing OOP concepts such as inheritance and polymorphism to manage food items.

Data Analysis & Visualization: Using the Matplotlib library to visualize data from text files or integrated Pandas or NumPy modules to plot histograms, scatter plots, and bar graphs in analyzing the sales data.

i. **Input Specification:**

The program will require input for creating the menu items (name, category, price) and customer order (selected items and quantities). The program will also require input for the file paths for reading the existing data.

- **Output Specification:**

The system will generate the calculated bills for the customers and present sales reports that summarize popular items and category revenues. It will also generate graphs and visual plots to present daily or monthly sales revenues and item category sales distribution.

- **Constraints and Assumptions:**

The software operates using local files and not a relational database to store the data permanently in text files.
Input validation: The quantity input and item selection must be valid, and the program will assume that the user may provide invalid inputs and thus implement exception handling.

- **Learning Outcomes:**

After completing the project, we will master the concepts of Python Object-Oriented Programming, advanced error-handling techniques such as exception handling, and file-handling functions. We will also gain experience in using the Matplotlib library in generating visual plots from text files to represent business insights.

- **Platform Required:**

Jupyter Notebook, Visual Studio code, or Python IDE with the Matplotlib library.

- **Books and Link Sources:**

1. Let Us Python (5th Edition) by Yashvant Kanetkar.
2. Python Crash Course: A Hands-On, Project-Based Introduction to Programming (3rd Edition) by Eric Matthes.
3. Python Programming: Using Problem Solving Approach (3rd Edition) by Reema Thareja.
4. *Corey Schafer's Python Tutorials* (YouTube Channel) - For clear, step-by-step video tutorials covering core Python concepts and file handling.